國立陽明交通大學

資訊管理研究所

碩士論文

Institute of Information Management

National Yang Ming Chiao Tung University

Master Thesis

基於理由感知門控與稀疏評論面向知識圖譜之產品推薦

Product Recommendation via Rationale-Aware Gating over Sparse Review-Aspect Knowledge Graphs

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中華民國 一一五年六月

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A Thesis
Submitted to Institute of Information Management
College of Management
National Yang Ming Chiao Tung University
in partial Fulfillment of the Requirements
for the Degree of
Master of Science
in
Information Management

June 2026
Taiwan, Republic of China

中華民國 一一五年六月

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摘　要

知識圖譜感知推薦在稠密且品質良好的圖譜上通常能提升準確度；然而，在評論衍生、冷啟動或隱私受限等情境中，圖譜往往稀疏或品質不穩，相關研究仍相對不足。本文以商品評論衍生的稀疏圖譜為例，過濾後每件商品平均僅有 2.4 條面向邊；實驗顯示，代表性的知識圖譜感知方法皆不如完全不使用圖譜的純協同過濾模型。主要原因在於，這些方法將圖譜訊號硬性嵌入評分流程，缺乏在圖譜不可靠時自動降低其影響的機制，使雜訊干擾協同訊號的學習。

本文提出 RA-GARK，將圖譜視為一條可調控的側通道，而非評分流程中必須使用的組件，並透過三個步驟實現。其一，將每件商品的圖譜內容整理為少數代表性面向，並以帶有語意的向量初始化，避免完全隨機的起點。其二，模型會依使用者與商品，自動選取最相關的面向。其三，由融合閘決定是否啟用此側通道：訓練初期幾乎關閉，使模型從純協同過濾出發；只有當側通道確實有幫助時才逐步打開，並在其無用時平順退回。在不更動圖譜本身的前提下，RA-GARK 將圖譜的貢獻由負面轉為正面。

本研究指出，知識圖譜感知方法在稀疏圖譜上表現不佳，關鍵並非圖譜完全沒有資訊，而是既有架構缺乏選擇性使用圖譜的能力。同一個圖譜在強制融合下可能成為負擔，但當模型能自行決定是否採用時，便可轉為助益。這種優雅退化的設計並不限於知識圖譜；對於任何需要結合可靠主訊號與品質不穩定輔助訊號的情境，例如冷啟動、多模態或跨領域推薦，同樣具有應用價值。

關鍵字：知識圖譜推薦；可退化機制；融合閘；稀疏知識圖譜；協同過濾

Product Recommendation via Rationale-Aware Gating over Sparse Review-Aspect Knowledge Graphs

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Abstract

Knowledge-graph-aware recommenders often improve accuracy when the KG is dense and reliable. However, in review-derived, cold-start, or privacy-constrained settings, KGs are often sparse or noisy, and this regime remains underexplored. Using a sparse review-derived product KG with only 2.4 aspect edges per item after filtering, we observe that representative KG-aware methods are all outperformed by a plain collaborative-filtering model that ignores the KG entirely. The main reason is that these methods hardwire the KG signal into the scoring pipeline, with no mechanism to reduce its influence when the KG is unreliable, allowing KG noise to interfere with collaborative-signal learning.

We propose RA-GARK, which treats the KG as a controllable side channel rather than a mandatory component of the scoring process, realized in three steps. First, each item's graph content is summarized into a few representative aspects and initialized from semantic vectors instead of purely random values. Second, the model automatically selects the most relevant aspect for each user and item. Third, a fusion gate decides whether to use this side channel: it starts almost closed, so training begins from pure collaborative filtering, and opens only when the side channel is useful, allowing the model to fall back gracefully when it is not. Without altering the KG itself, RA-GARK turns the KG's contribution from negative to positive.

This work shows that the poor performance of KG-aware recommenders on sparse KGs is not because the KG contains no useful information, but because existing architectures lack the ability to use the KG selectively. The same KG can be a liability under mandatory fusion, yet become an asset once the model can decide whether to use it. This kind of graceful degradation is not specific to knowledge graphs; it is also valuable in any setting that combines a reliable primary signal with an auxiliary signal of uncertain quality, such as cold-start, multimodal, or cross-domain recommendation.

Keywords: knowledge-graph-aware recommendation; graceful degradation; fusion gate; sparse knowledge graph; collaborative filtering.

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Chapter 1. Introduction

1.1 Background

Graph neural networks (GNNs) have become the dominant model family for collaborative filtering. Among them, LightGCN [1] demonstrated that removing the non-linear activation and feature transformation from a GCN, and retaining only linear neighborhood aggregation, is sufficient to surpass deeper alternatives such as NGCF [2] on standard recommendation benchmarks. The simplicity and stability of LightGCN have since established it as a strong, parameter-efficient backbone, on top of which more recent methods such as SGL [3] layer additional self-supervised objectives.

A complementary line of research, knowledge-graph-aware recommendation, augments collaborative signals with item-side semantics encoded in a knowledge graph (KG). Representative methods include KGAT [4], which propagates messages over a unified user–item–entity graph; KGCL [5] and MCCLK [6], which align KG-derived views with collaborative views via contrastive learning; and KGRec [7], which introduces edge-level rationalization to identify which KG triples are responsible for a recommendation. When the KG is dense and high-quality, these methods consistently outperform pure collaborative baselines.

1.2 An Empirical Anomaly under Sparse KG

Our investigation begins with an empirical observation that motivates the rest of this thesis. On the review-derived KG dataset used throughout this work, an Amazon Books subset with on average 2.4 aspect edges per item after filtering, all four representative KG-aware methods listed above are outperformed by pure LightGCN that does not consult the KG at all. Table 1 summarizes the gap.

This is not a claim that the four baselines are flawed, each performs strongly on the dense KG datasets reported in their original papers. Rather, the observation suggests a failure mode shared by deep KG–CF coupling: when the KG is too sparse or noisy to provide reliable item

Table 1: NDCG@20 of representative KG-aware methods on our sparse review-derived KG, contrasted with pure LightGCN. All four KG-aware methods underperform the no-KG baseline.

Method	NDCG@20
MCCLK [6]	0.1067
KGCL [5]	0.1073
KGAT [4]	0.1079
KGRec [7]	0.1095
Pure LightGCN [1] (no KG)	**0.1179**

semantics, fusing it directly into the scoring path drags KG noise into the gradient flow that would otherwise produce a clean collaborative signal.

1.3 Sparse KG as a Real-World Default

A natural objection is to treat the empirical anomaly as a dataset peculiarity and either curate a denser KG, or apply KG completion to fill in missing edges. We argue that sparse KG is in fact the more common setting in practical recommendation, and that robustness under unreliable KG signal therefore deserves dedicated study rather than circumvention.

Three sources of sparsity recur in practice. First, *review-derived KGs*, of the kind used in this work, inherit their density from whatever topics users happened to mention; coverage is inherently uneven. Second, *cold-start and emerging domains* (new product categories, low-resource languages, niche verticals) rarely have access to a curated source comparable to Freebase or Wikidata, so the KG must be bootstrapped from limited signal. Third, *privacy-constrained domains* such as medical or financial recommendation deliberately restrict which relational signals can be exposed to a model. In all three settings, an aggressive KG completion step is itself non-trivial: completion introduces its own noise, and typically requires sufficient seed signal to train, which is precisely what is missing in the sparse setting.

The implication is that a recommender's *robustness* when the KG is unreliable is at least as important as its *peak performance* when the KG is dense. This thesis takes that robustness as the primary design objective.

1.4 Research Questions

The empirical anomaly and the practical importance of the sparse-KG setting motivate two research questions:

- **(RQ1, diagnosis)** Why do existing KG-aware methods underperform pure LightGCN under sparse KG? What is the structural failure mode they share?
- **(RQ2, prescription)** What design principle allows a model to exploit the KG when its signal is informative, while preventing KG noise from contaminating collaborative filtering when it is not?

The thesis statement of this work is that the KG should not be a mandatory component of the scoring pipeline, as it effectively is in KGAT-style deep fusion, but should instead be a side channel that can be *explicitly gated* based on whether its signal is helpful. Concretely, this idea is realized by three design choices: a softmax aspect-saliency attention that selects KG rationales per user–item pair; a KG-driven SVD initialization that preserves semantic geometry; and a local-biased fusion gate whose initialization implements graceful degradation to pure LightGCN when the KG turns out to be uninformative.

1.5 Contributions

This thesis makes the following contributions:

1. **An empirical study** demonstrating that, under sparse review-derived KG, four representative KG-aware methods (KGAT, KGCL, MCCLK, KGRec) are uniformly outperformed by pure LightGCN. We attribute this to the absence of an architectural mechanism that can disengage the KG when it is unreliable.
2. **RA-GARK, a dual-view architecture** that integrates collaborative and KG-derived signals through a side-channel design rather than deep fusion. RA-GARK consists of three core ingredients, each of which is independently necessary:
   - **Softmax aspect-saliency attention** (Section 3.7). Per-item KG semantics are stored across $A = 4$ *latent* aspect slots produced by a truncated SVD of an IDF-weighted item–aspect co-occurrence matrix; the slot count is a model hyperparameter and

does not correspond to a fixed number of raw aspect tags per item. A softmax-normalized attention selects, per user–item pair, which latent slot best represents the item. Removing softmax in favor of sigmoid degrades NDCG@20 by 15.5%, the largest single-component effect we observe.

- **Local-biased fusion gate** (Section 3.8). The fusion gate is initialized with a positive bias of $+5$, so that $\alpha^{(0)} \approx 0.993$. The model thus begins training as essentially pure LightGCN, and only opens the KG channel when gradients indicate it helps. This provides a structural *graceful degradation* guarantee absent from the four baselines. Removing the bias init costs 4.5% NDCG@20.

- **KG-SVD initialization** (Section 3.6). The aspect-slot embeddings are initialized from a truncated SVD of an IDF-weighted item–aspect matrix, preserving the KG's semantic geometry as the optimization starting point. Replacing this with random initialization costs 5.9% NDCG@20.

3. **Empirical validation** on the sparse review-derived KG benchmark, where RA-GARK reaches NDCG@20 $= 0.1242$, exceeding the strongest KG-aware baseline (KGRec) by $+13.4\%$ and, more importantly, exceeding pure LightGCN by $+5.3\%$. The latter gap demonstrates that the proposed design principle flips the contribution of KG signal from *net-negative* (as in the four baselines) to *net-positive*, even when the KG itself is unchanged.

4. **A methodological observation** regarding attention normalization in rationale-aware design. The unusually large effect of softmax versus sigmoid in our setting suggests that the choice of normalization should be aligned with the semantic assumption of the rationale. We discuss this as a single-dataset hypothesis rather than a general claim, pending replication on additional benchmarks (Chapter 5).

1.6 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 reviews related work along four axes: graph collaborative filtering, KG-aware recommendation, gating mechanisms in deep learning, and attention normalization. Chapter 3 presents the RA-GARK architecture in full: it states the design principle, fixes notation and the problem formulation, and develops each module, the LightGCN local view, the global KG view with its two innovations (KG-SVD initialization and softmax aspect-saliency attention), the local-biased fusion gate, and the cross-view contrastive regularization, together with the training procedure and computational complexity. Chapter 4 reports experimental results, ablations, sensitivity analyses, stability of results, and a case study. Chapter 5 discusses the methodological insight on attention normalization, acknowledges the limitations of single-dataset validation, and outlines future work.

Chapter 2. Related Work

This chapter reviews prior work along four axes that together motivate the design of RA-GARK. Section 2.1 covers the two foundations on which our local and global views are positioned: LightGCN as the collaborative-filtering backbone and KGAT as the canonical deep KG–CF fusion. Section 2.2 reviews contrastive KG methods that align collaborative and KG views without explicit gating. Section 2.3 discusses KGRec, the rationale-aware method most closely related to ours, and Section 2.4 contrasts our rationale paradigm with KGRec's under the lens of *KG trust*. Sections 2.5 and 2.6 survey gating mechanisms in deep learning broadly and observe that no existing KG-aware method employs a fusion gate to provide architectural graceful degradation. Section 2.8 summarizes how RA-GARK is positioned with respect to all of the above.

2.1 Foundations: LightGCN and KGAT

LightGCN [1] is the immediate predecessor of our local view. Building on the observation that the non-linear feature transformation in NGCF [2] contributes little to recommendation accuracy, LightGCN strips a graph convolutional network down to its bare essentials: a normalized neighborhood aggregation $\mathbf{E}^{(\ell+1)} = \tilde{\mathbf{A}}\mathbf{E}^{(\ell)}$ over the user–item bipartite graph, with the final embedding obtained as the layer-wise average $\bar{\mathbf{E}} = \frac{1}{K+1}\sum_{\ell=0}^{K}\mathbf{E}^{(\ell)}$. The simplification consistently outperforms NGCF on standard recommendation benchmarks while being faster and more stable to train. Subsequent self-supervised methods such as SGL [3] and DCCF [8] layer additional contrastive objectives on top of a LightGCN-style backbone.

In RA-GARK, we adopt LightGCN as the local view *verbatim* (with $K = 2$): no KG-related operation is applied within this branch. The motivation is twofold. First, on our sparse review-derived KG, pure LightGCN already achieves NDCG@20 = 0.1179, surpassing all KG-aware baselines we evaluated. Any deviation from this strong default must therefore demonstrably improve, not degrade, the result, a hard design constraint we revisit throughout this thesis. Second, isolating the KG signal to a separate global view (Chapter 3) prevents KG noise from contaminating the gradient flow that produces the collaborative representation.

KGAT [4] represents the canonical *deep fusion* approach to KG-aware recommendation. KGAT merges the user–item interaction graph with a knowledge graph into a single Collaborative Knowledge Graph (CKG), and propagates messages over this unified structure with a bi-interaction aggregator. KG entity embeddings thus participate directly in the formation of user and item representations. When the KG is dense and high-quality, this design yields strong improvements over CF baselines.

The implicit assumption is that the KG can be trusted to inject useful signal everywhere it appears in the propagation path. In our sparse setting this assumption breaks down: KGAT achieves only NDCG@20 = 0.1079, below pure LightGCN. Subsequent work [5, 6, 7] retains the KGAT-style aggregator and inherits the same vulnerability. RA-GARK reverses this design choice: we do not merge the KG and user–item graphs at all, and instead route KG signal through a dedicated side channel that the model can attenuate or fully disengage when the KG proves unreliable.

2.2 Contrastive Knowledge-Graph Methods

KGCL [5] introduces relation-level structural perturbations to the KG and aligns the perturbed and original views via an InfoNCE [9] contrastive objective. The intuition is that informative KG structure should be invariant under stochastic perturbation, while noisy edges should not. KGCL substantially outperforms KGAT on dense KGs. In our sparse setting, however, the perturbed KG retains too few edges for the contrastive objective to provide reliable supervision: NDCG@20 drops to 0.1073.

MCCLK [6] extends contrastive alignment to three views simultaneously (collaborative, semantic, and structural) with pairwise contrastive losses among all three. While this multi-view design achieves state-of-the-art on dense KG benchmarks, it inherits the same dense-KG assumption: under sparse KG, two of the three views (semantic and structural) carry insufficient signal, and the contrastive alignment becomes a regularizer dominated by noise rather than structure (0.1067 NDCG@20).

RA-GARK also incorporates cross-view contrastive learning, but in a deliberately limited role. Our contrastive losses ($\mathcal{L}_{\mathrm{aCL}}$ and $\mathcal{L}_{\mathrm{uCL}}$, Section 3.9) act only as auxiliary geometric regularizers with a small weight ($\lambda = 0.005$), and crucially apply a stop-gradient on the KG side so that the contrastive gradient never propagates back into the SVD-initialized aspect embeddings.

Compared to KGCL/MCCLK, contrastive learning here is a cosmetic alignment layer, not the primary fusion mechanism; the actual integration of collaborative and KG signals is handled by an explicit gate (Section 2.5).

2.3 Rationale-Aware Recommendation: KGRec

KGRec [7] is the work most directly related to ours. It is the first KG-aware recommender to make rationale selection explicit: instead of treating every KG triple as equally informative, KGRec computes an attention-based importance score per KG edge, and uses Bernoulli dropout to stochastically remove high-attention edges, encouraging the model not to depend on a small set of dominant edges. Robustness is then enforced via a contrastive objective between the original and dropped graphs. The rationalization machinery is built on top of a KGAT-style aggregator, so the rationale operates within the message-passing path of a unified user–item–entity graph. On our sparse benchmark KGRec obtains NDCG@20 = 0.1095, the strongest among the four KG-aware baselines but still below pure LightGCN.

RA-GARK adopts the rationale-aware paradigm in spirit but differs from KGRec in several structural choices that we elaborate next.

2.4 Rationale Paradigm: Shared Premise vs. Diverged Trust Assumption

KGRec [7] and RA-GARK both endorse the observation that not all KG edges contribute equally to a recommendation, and that an explicit selection mechanism is needed. They differ, however, in what they assume about the *trustworthiness of the KG as a whole*, and this assumption propagates into divergent design choices.

KGRec's implicit assumption is that informative edges *exist* within the KG, so the rationalization task reduces to finding them: stochastic dropout plus contrastive learning suffice to identify and amplify the useful subset. The KG itself is assumed to be reliable as a body of evidence; only individual edges are weighted.

RA-GARK's premise is weaker. We do not assume that the KG, taken as a whole, is reliable. The aspect-saliency softmax (Section 3.7) selects rationales when the KG is informative, but the local-biased fusion gate (Section 3.8) can additionally *disengage the entire KG channel* when the gate's gradient indicates it is not. This trust difference manifests in four design divergences:

- **Rationale granularity.** KGRec selects at the edge level (KG triples). RA-GARK aggregates each item's KG semantics into $A = 4$ *latent* aspect slots produced by a truncated SVD of the IDF-weighted item–aspect co-occurrence matrix, and selects at the slot level. The slot count is a model hyperparameter unrelated to the number of raw aspect tags any individual item carries; the granularity is therefore coarser than edge-level selection but produces directly interpretable per-item aspect preferences.

- **Selection mechanism.** KGRec uses discrete Bernoulli dropout combined with contrastive learning. RA-GARK uses a differentiable softmax attention, which is end-to-end trainable and exposes inference-time weights as an interpretability artifact.

- **Site of action.** KGRec applies its rationalization *within* the KGAT message-passing path; rationale and aggregation are entangled. RA-GARK applies rationalization in a side channel that is separate from the LightGCN local view, and only late-stage gating combines the two.

- **Treatment of KG trust.** KGRec has no architectural mechanism to disengage the KG. RA-GARK's bias-initialized fusion gate provides exactly this mechanism: at the start of training, $\alpha^{(0)} \approx 0.993$, so the model is essentially pure LightGCN; the gate opens only when learning indicates that the KG channel reduces loss.

Table 2 consolidates these four axes side by side.

Table 2: Rationale-paradigm comparison: KGRec vs. RA-GARK along the four design axes.

Axis	KGRec	RA-GARK
Rationale granularity	KG triple (edge)	A latent aspect slots
Selection mechanism	Bernoulli dropout + CL	Softmax attention
Site of action	Inside KGAT propagation	Side channel, late fusion
Treatment of KG trust	No disengage mechanism	Bias-initialized gate

We therefore position RA-GARK not as an engineering improvement of KGRec, but as a separate design that begins from a different premise about KG reliability. Whether this distinction matters empirically is settled in Chapter 4.

2.5 Gating Mechanisms in Deep Learning

The fusion gate in RA-GARK has two close analogues outside KG-aware recommendation, both of which inform our design.

Highway Networks [10] introduce a learnable gate $T(\mathbf{x})$ that interpolates between a transformed signal $H(\mathbf{x})$ and an identity skip: $\mathbf{y} = T(\mathbf{x}) \cdot H(\mathbf{x}) + (1 - T(\mathbf{x})) \cdot \mathbf{x}$. The pivotal design choice is the negative bias initialization on T: at the start of training, $T \approx 0$ and the network behaves as the identity map. Subsequent training opens the gate only where the transformation is beneficial. This was originally motivated by the difficulty of training very deep networks, but the structural pattern, bias the gate toward a known-safe default and let learning override the default only when justified, generalizes well beyond depth. RA-GARK transplants this pattern from in-network skip to dual-view fusion: our bias of $+5$ on the fusion gate plays the same role as Highway's negative bias, yielding a graceful-degradation guarantee with respect to the KG channel.

Mixture-of-Experts gating, exemplified by MMoE [11] and PLE [12] in multi-task recommendation, employs a gating network to weight multiple expert towers. While conceptually similar to multi-pipeline selection, two important properties differ from RA-GARK. First, the experts in MMoE/PLE are *homogeneous candidates*, instances of the same architecture trained on different task-specific objectives, so there is no notion of one expert being intrinsically less reliable than another. Second, the gates in these architectures use symmetric initialization (no bias toward any particular expert at training start), which is appropriate for the symmetric expert setting but not for our asymmetric setting in which one channel (LightGCN) is known to be a strong default and another (KG) is known to be sometimes uninformative.

A further line of work, contrastive dual-view recommenders such as SGL [3] and DCCF [8], builds multiple views by perturbing a single graph (edge dropout, node dropout, random walks). The two views in such methods reside in the *same signal space*, and contrastive alignment is sufficient to integrate them. RA-GARK's two views, in contrast, are structurally heterogeneous: a user–item bipartite graph for the local view and an item–aspect KG for the global view. Implicit alignment via contrastive learning alone cannot adjust the relative weight of two heterogeneous signal spaces; an explicit gate is needed.

2.6 Gap: No Fusion Gate in KG-Aware Recommendation

Despite the maturity of gating in deep learning, no method in the KG-aware recommendation literature we surveyed employs a fusion gate as the integration mechanism between collaborative and KG signals. Table 3 summarizes the integration mechanisms used by the four benchmarked baselines and three additional well-known KG-aware methods (CKAN [13], MKR [14], KGIN [15]). Across this set, KG signals are integrated via direct entity propagation, contrastive alignment, edge-level dropout, attention, cross-feature units, or intent-disentanglement modules. None of these mechanisms allows the KG channel to be *architecturally disengaged* when its signal is unreliable.

Table 3: Integration mechanisms used by representative KG-aware recommenders. None provide architectural graceful degradation under unreliable KG.

Method	KG integration mechanism	Disengageable?
KGAT [4]	Direct entity-message propagation	No
KGCL [5]	Perturbed-KG contrastive learning	No
MCCLK [6]	Multi-view contrastive alignment	No
KGRec [7]	Edge-level dropout + KGAT aggregator	No
CKAN [13]	Collaborative-knowledge attention	No
MKR [14]	Cross-feature transfer unit	No
KGIN [15]	Intent disentanglement over KG	No
RA-GARK (this work)	**Bias-initialized fusion gate**	**Yes**

We make a deliberately narrow claim. We do not claim that RA-GARK is the first to use a gate of any kind in KG-aware recommendation, since the KG-aware literature is large and exhaustive enumeration is infeasible. The defensible claim is the following: *within the methods we examined, none uses a bias-initialized fusion gate to provide architectural graceful degradation under sparse or unreliable KG.* Our ablation in Chapter 4 attributes a 4.5% NDCG@20 contribution specifically to this gate's bias initialization, supporting the claim that the design choice is non-trivial in this setting.

2.7 Attention Normalization

A relevant cross-cutting question is which normalization, sigmoid or softmax, should be applied to attention weights when selecting rationales. The literature does not converge on a single answer. DIN [16] applies a sigmoid attention over user behavior history under the assumption that each historical behavior is independently relevant to a candidate. NAIS [17] applies softmax attention over historical interactions for item similarity, and AFM [18] uses softmax over feature interactions. The choice in each work is justified by the semantic assumption underlying the attention.

In Chapter 4 we report a single-dataset observation that, for our aspect-saliency rationale, replacing softmax with sigmoid costs 15.5% NDCG@20, the largest single-component effect we measure. We interpret this as supporting the design heuristic that normalization should match the semantic assumption of the rationale, but explicitly frame the observation as a hypothesis pending multi-dataset replication (Chapter 5).

2.8 Summary and Positioning of RA-GARK

Table 4 consolidates the comparison along four design axes. Among the methods evaluated on our sparse benchmark, RA-GARK is the only one that combines a pure-LightGCN local aggregator with a gated late-fusion of the KG signal, and the only one whose rationale operates at the item-aspect granularity rather than the KG-edge granularity.

Table 4: Design comparison along four axes. NDCG@20 is reported on our sparse review-derived KG benchmark.

Method	Local aggregator	KG integration	Rationale	NDCG@20
KGAT	Bi-interaction	Direct propagation	None	0.1079
KGCL	KGAT-style	Propagation + CL	None	0.1073
MCCLK	Multi-view	Multi-view CL	None	0.1067
KGRec	KGAT-style	Propagation + edge dropout	Edge Bernoulli	0.1095
RA-GARK	**Pure LightGCN**	**Gated late fusion**	**Aspect softmax**	**0.1242**

We do not position RA-GARK as a strict improvement over the four baselines. Rather, RA-GARK occupies a distinct point in design space, one defined by treating the KG as an explicitly gateable side channel rather than a mandatory pipeline component. This choice yields strong results in our sparse-KG setting, and the gating-based graceful-degradation property may also be useful when the KG is of uneven quality or when the model is deployed across domains of differing KG density. When the KG is dense and uniformly reliable, on the other hand, deep fusion approaches such as KGAT or KGRec may still be preferable, since the late-fusion design forgoes some of the representational expressiveness available to deep-fusion architectures.

Chapter 3. Methodology

This chapter presents RA-GARK in full. The chapter is organized as follows:

- Section 3.1 states the central design principle that distinguishes RA-GARK from the deep-fusion baselines of Chapter 2.
- Section 3.2 gives a swim-lane overview of the architecture and inference-time data flow.
- Sections 3.3–3.4 fix notation and formalize the recommendation task and training objective.
- Sections 3.5–3.9 detail each module in the order it participates in the forward pass.
- Section 3.10 describes training and computational complexity.

3.1 Design Principle: KG as a Gateable Side Channel

The empirical observation in Chapter 1 is that four representative KG-aware methods (KGAT, KGCL, MCCLK, KGRec) are uniformly outperformed by pure LightGCN on our sparse review-derived knowledge graph (KG).

The underlying cause shared by these baselines is that the KG is wired into the scoring pipeline as a *mandatory* component: KG entity embeddings participate in message passing, and there is no architectural switch by which the model can disengage the KG when it turns out to be unreliable. KG noise therefore enters the same gradient flow that produces the collaborative representation.

RA-GARK is organized around a single design principle that addresses this failure mode directly:

> *The KG should not be a mandatory component of the scoring pipeline. It should be a side channel whose contribution to the final score is determined by a learned gate, biased at initialization toward a safe default (pure collaborative filtering) and opened only when training signal indicates that the KG channel reduces loss.*

Three design consequences follow from this principle and define RA-GARK:

1. **Two parallel views.** A *local view* produces a purely collaborative representation $(u_{\mathrm{loc}}, i_{\mathrm{loc}})$ via LightGCN over the user–item interaction graph, with no KG involvement. A *global view* produces a KG-derived representation $(u_{\mathrm{glo}}, i_{\mathrm{glo}})$ from a separate user embedding table and the per-item latent aspect slots, with no collaborative signal involvement. The two views remain structurally disjoint up to the fusion stage.

2. **Late, gated fusion.** The two views are combined only at the final scoring stage by a fusion gate that produces a per-(user, item) mixing weight $\alpha \in (0, 1)$. The gate is bias-initialized so that $\alpha^{(0)} \approx 0.993$, placing the model essentially at pure LightGCN at training start. The KG channel is opened only insofar as gradients drive α downward during training.

3. **Rationale-aware item-side aggregation.** Each item's KG semantics are stored in $A = 4$ *latent* aspect slots produced by truncated SVD of an IDF-weighted item–aspect matrix; a softmax aspect-saliency attention selects, per (user, item) pair, which latent slot represents the item in the global view.

Each consequence is mathematized in Sections 3.5–3.9 below. The remainder of this chapter first fixes notation and the problem statement (Sections 3.3–3.4), then develops each module in the order it participates in the forward pass.

3.2 Architectural Overview

Figure 1 sketches the data flow at inference time. The model is laid out as two horizontal lanes corresponding to the local and global views, joined at the right by the fusion stage. The local view (top lane) applies LightGCN to the user–item interaction graph to obtain $(u_{\mathrm{loc}}, i_{\mathrm{loc}})$. The global view (bottom lane) reads the user's global embedding u_{glo} by direct lookup, and obtains i_{glo} by applying softmax rationale masking over the $A = 4$ latent aspect slots of the item; the slots themselves are SVD-initialized (indicated by the dashed annotation in the figure). The two views are mixed by independent user-side and item-side fusion gates, each bias-initialized to $+5$, and the final score is the inner product $\langle u_{\mathrm{final}}, i_{\mathrm{final}} \rangle$. The cross-view contrastive paths are omitted from the figure for clarity and are detailed in Section 3.9.

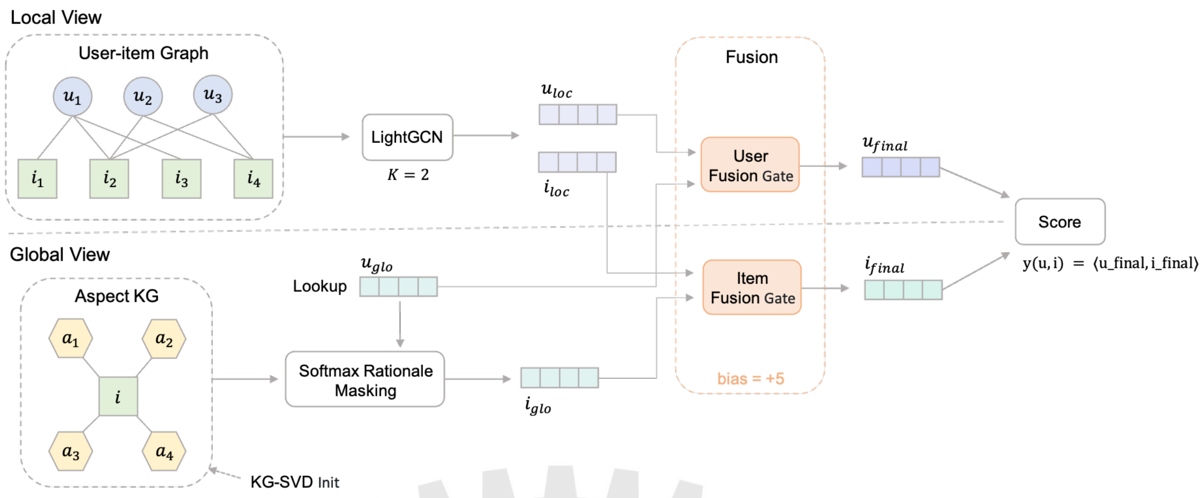

Figure 1: RA-GARK architecture.

Two design properties are worth highlighting before the formal presentation. First, u_{glo} does not pass through the rationale masking module: it is obtained by a direct embedding lookup, and feeds straight into the user-side fusion gate. The rationale masking applies only on the item side, consistent with the semantic that the rationale answers "which aspect of *this item* matters to *this user*." Second, the KG-SVD step is an initialization, not a forward-pass operation: the SVD is computed once before training and used to set the values of the aspect-slot parameters, after which the slots are updated by gradient descent like any other parameter.

3.3 Notation

Throughout the remainder of the thesis we use the symbols listed in Table 5.

3.4 Problem Formulation

Task. We address top-K recommendation in the implicit-feedback setting. Given $\mathcal{U}$, $\mathcal{I}$, observed interactions $\mathcal{R}$, and the item–aspect KG $\mathcal{G}_{\text{KG}}$, the goal is to produce for each user $u \in \mathcal{U}$ a ranked list of items from $\mathcal{I} \setminus \{i : (u, i) \in \mathcal{R}\}$, evaluated by ranking metrics (NDCG@K, Recall@K, Hit@K) against a held-out test interaction set.

Scoring function. RA-GARK assigns to each (u, i) pair the inner-product score

$$\hat{y}(u, i) = \langle u_{\text{final}},\ i_{\text{final}} \rangle, \tag{3.1}$$

Table 5: Notation used in this and subsequent chapters.

Symbol	Meaning
$\mathcal{U}, \mathcal{I}$	sets of users and items
$\mathcal{R} \subseteq \mathcal{U} \times \mathcal{I}$	observed user–item interactions
$\mathcal{G}_{\mathrm{KG}}$	item–aspect knowledge graph (review-derived)
$\mathcal{A}$	set of unique aspect tags appearing in $\mathcal{G}_{\mathrm{KG}}$
A	number of latent aspect slots per item ($A = 4$)
d	embedding dimension ($d = 128$ unless stated otherwise)
K	number of LightGCN propagation layers ($K = 2$)
$u_{\mathrm{loc}}, i_{\mathrm{loc}} \in \mathbb{R}^d$	local-view (LightGCN) embeddings
$u_{\mathrm{glo}} \in \mathbb{R}^d$	global-view user embedding (direct lookup)
$\mathbf{a}_i \in \mathbb{R}^{A \times d}$	latent aspect slots of item i (SVD-initialized)
$i_{\mathrm{glo}} \in \mathbb{R}^d$	global-view item embedding (rationale-masked)
$\alpha_u, \alpha_i \in (0, 1)$	user- and item-side fusion gate outputs
$\hat{y}(u, i) \in \mathbb{R}$	predicted preference score for pair (u, i)

where u_{final} and i_{final} are produced by gated fusion of the local and global representations:

$$u_{\mathrm{final}} = \alpha_u \, u_{\mathrm{loc}} + (1 - \alpha_u) \, u_{\mathrm{glo}}, \qquad i_{\mathrm{final}} = \alpha_i \, i_{\mathrm{loc}} + (1 - \alpha_i) \, i_{\mathrm{glo}}. \tag{3.2}$$

The gate outputs α_u, α_i are computed by independent MLPs (Section 3.8). With the bias initialization used throughout this thesis, $\alpha_u^{(0)}, \alpha_i^{(0)} \approx 0.993$, so $\hat{y}^{(0)}(u, i) \approx \langle u_{\mathrm{loc}}, i_{\mathrm{loc}} \rangle$ at the start of training, i.e. the score is essentially that of pure LightGCN until the gate learns otherwise.

Training objective. We optimize a Bayesian Personalized Ranking (BPR) [19] loss with two auxiliary cross-view contrastive regularizers:

$$\mathcal{L} = \mathcal{L}_{\mathrm{BPR}} + \lambda_{\mathrm{CL}} \big(\mathcal{L}_{\mathrm{aCL}} + \mathcal{L}_{\mathrm{uCL}} \big), \tag{3.3}$$

where for each observed pair $(u, i^+) \in \mathcal{R}$ and a sampled negative item $i^- \notin \mathcal{R}_u$,

$$\mathcal{L}_{\mathrm{BPR}} = -\log \sigma \big(\hat{y}(u, i^+) - \hat{y}(u, i^-) \big), \tag{3.4}$$

and $\mathcal{L}_{\mathrm{aCL}}, \mathcal{L}_{\mathrm{uCL}}$ are InfoNCE-style alignment losses between $(i_{\mathrm{loc}}, i_{\mathrm{glo}})$ and $(u_{\mathrm{loc}}, u_{\mathrm{glo}})$ respectively, with a stop-gradient applied on the global side to prevent the contrastive gradient from disturbing the SVD-initialized aspect slots (Section 3.9). The contrastive weight λ_{CL} is fixed at 0.005, keeping the contrastive role auxiliary rather than primary.

3.5 Local View: LightGCN

The local view produces a purely collaborative representation $(u_{\text{loc}}, i_{\text{loc}})$ from the user–item interaction graph alone. No KG operation is applied within this branch. The architecture is identical to LightGCN [1].

3.5.1 Motivation

LightGCN is chosen as the local-view backbone for three reasons. First, on the sparse review-derived KG used in this thesis, pure LightGCN already exceeds all four KG-aware baselines we evaluated (Chapter 1), making it the strongest non-KG starting point against which any subsequent design must demonstrably improve. Second, its minimal parameterization (only the user and item embedding tables, no learned weights in the propagation) limits the surface area through which KG noise could leak into the collaborative gradient flow when the two views are coupled later. Third, the propagation is purely linear, so each layer's contribution to the final representation can be read off the closed-form layer-wise average, and design decisions about gate placement (Section 3.8) and contrastive alignment (Section 3.9) can be reasoned about without nonlinearity confounds.

3.5.2 Graph Construction

Let $\mathbf{R} \in \{0,1\}^{|\mathcal{U}|\times|\mathcal{I}|}$ be the user–item interaction matrix derived from $\mathcal{R}$, with $\mathbf{R}_{u,i} = 1$ if $(u,i) \in \mathcal{R}$ and 0 otherwise. We form the bipartite adjacency

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{R} \\ \mathbf{R}^\top & \mathbf{0} \end{bmatrix} \in \mathbb{R}^{(|\mathcal{U}|+|\mathcal{I}|)\times(|\mathcal{U}|+|\mathcal{I}|)}, \tag{3.5}$$

and its symmetrically normalized variant

$$\tilde{\mathbf{A}} = \mathbf{D}^{-1/2}\,\mathbf{A}\,\mathbf{D}^{-1/2}, \qquad \mathbf{D} = \operatorname{diag}(\mathbf{A}\,\mathbf{1})\,. \tag{3.6}$$

The normalization downweights edges incident to high-degree users or popular items, which empirically stabilizes training on long-tailed interaction distributions.

3.5.3 Propagation

Let $\mathbf{E}^{(0)} \in \mathbb{R}^{(|\mathcal{U}|+|\mathcal{I}|)\times d}$ be the initial embedding table, with the first $|\mathcal{U}|$ rows storing user embeddings and the remaining $|\mathcal{I}|$ rows storing item embeddings. Both blocks are initialized with Xavier-normal random entries. The ℓ-th layer is obtained by a single normalized neighbor aggregation:

$$\mathbf{E}^{(\ell+1)} = \tilde{\mathbf{A}}\,\mathbf{E}^{(\ell)}, \qquad \ell = 0, 1, \ldots, K-1. \tag{3.7}$$

There is no learnable weight matrix, no nonlinearity, and no self-loop bias, the only learnable parameters in the entire local view are the $|\mathcal{U}|d + |\mathcal{I}|d$ entries of $\mathbf{E}^{(0)}$.

3.5.4 Layer-wise Aggregation and Output

We adopt the standard LightGCN layer-wise average

$$\bar{\mathbf{E}} = \frac{1}{K+1}\sum_{\ell=0}^{K} \mathbf{E}^{(\ell)}, \tag{3.8}$$

and read off u_{loc} and i_{loc} from the corresponding row of $\bar{\mathbf{E}}$. We use $K = 2$ throughout, consistent with the original LightGCN paper and with the depth at which our local-only validation loss plateaus.

The result is a pair of embeddings whose only information source is the user–item interaction graph. The local view participates in the final score $\hat{y}(u, i)$ only through the fusion stage of Section 3.8; no KG signal is allowed to alter u_{loc} or i_{loc} during training.

3.6 Global View: KG-SVD Initialization

The global view introduces a separate user-side embedding u_{glo} (a direct lookup, treated below) and an item-side parameter tensor that stores each item's KG semantics. The item-side tensor is the only place in RA-GARK where KG content enters the model. This section describes how the tensor is parameterized and how its values are initialized. Figure 2 gives a visual overview of the four-stage initialization pipeline; the remainder of this section formalizes each stage.

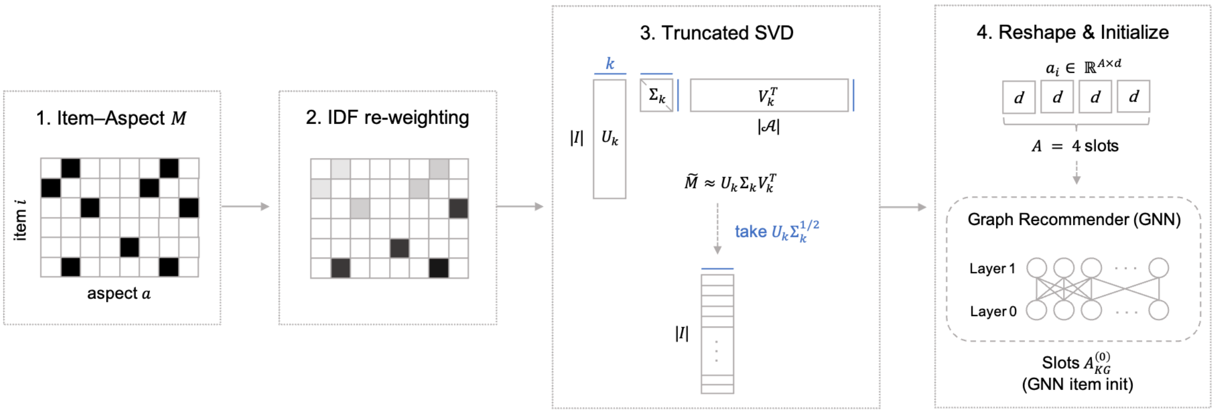

Figure 2: KG-SVD initialization pipeline.

3.6.1 Aspect Slots as the Item-side KG Representation

Each item i is represented in the global view by an aspect-slot tensor

$$\mathbf{a}_i \in \mathbb{R}^{A \times d}, \qquad A = 4, \tag{3.9}$$

which we collect for all items into a single learnable parameter $\mathbf{A}_{\text{KG}} \in \mathbb{R}^{|\mathcal{I}| \times A \times d}$. After initialization (described next), $\mathbf{A}_{\text{KG}}$ is updated by gradient descent like any other parameter.

A key clarification is in order: the slot count $A = 4$ is a model hyperparameter and does *not* correspond to a fixed number of raw aspect tags per item. In the review-derived KG used here, the average item carries only 2.4 aspect edges, and the global aspect-tag vocabulary is much larger ($|\mathcal{A}| \gg A$). The four slots are *latent* directions inside an aspect-induced semantic subspace, learned across all items collectively rather than tied to specific tags.

3.6.2 Co-occurrence Matrix

We start from a binary item–aspect co-occurrence matrix $\mathbf{M} \in \{0,1\}^{|\mathcal{I}| \times |\mathcal{A}|}$:

$$\mathbf{M}_{i,a} = \begin{cases} 1, & (i,a) \in \mathcal{G}_{\text{KG}}, \\ 0, & \text{otherwise.} \end{cases} \tag{3.10}$$

Each row of $\mathbf{M}$ is therefore a sparse indicator vector: at most a few entries per item are nonzero, and the global density (after pruning, Section 4.1) corresponds to an average of 2.4 ones per row.

3.6.3 IDF Weighting

Some aspect tags appear across many items (e.g. “well-written”, “engaging”); others are highly specific (e.g. a niche genre or theme). Without correction, the SVD that follows would be dominated by the high-frequency tags, producing latent directions that are essentially popularity priors. We mitigate this with a standard inverse document frequency (IDF) reweighting:

$$\mathrm{idf}(a) = \log\left(\frac{|\mathcal{I}|}{|\{i : \mathbf{M}_{i,a} = 1\}| + 1}\right) + 1, \tag{3.11}$$

and rescale columns of $\mathbf{M}$ by their IDF:

$$\tilde{\mathbf{M}}_{i,a} = \mathrm{idf}(a) \cdot \mathbf{M}_{i,a}. \tag{3.12}$$

The $+1$ in the denominator avoids division by zero on aspects whose entire item support has been pruned; the $+1$ outside the logarithm prevents the IDF of a maximally frequent aspect from collapsing to zero. Both adjustments are standard.

3.6.4 Truncated SVD

We compute a rank-k truncated SVD of $\tilde{\mathbf{M}}$:

$$\tilde{\mathbf{M}} \;\approx\; \mathbf{U}_k\, \mathbf{\Sigma}_k\, \mathbf{V}_k^\top, \qquad \mathbf{U}_k \in \mathbb{R}^{|\mathcal{I}| \times k}, \tag{3.13}$$

with $k = A \cdot d$. The left singular matrix $\mathbf{U}_k$ has one row per item; the rows live in a k-dimensional latent subspace within which IDF-weighted aspect co-occurrence is best explained at rank k.

We then form the per-item latent embedding by scaling each component by the square root of its singular value:

$$\mathbf{E}_{\mathrm{KG}} = \mathbf{U}_k\, \mathbf{\Sigma}_k^{1/2} \in \mathbb{R}^{|\mathcal{I}| \times k}. \tag{3.14}$$

The square-root scaling distributes the spectral energy evenly between the left and right factors, a standard choice that yields embeddings whose pairwise inner products approximate the IDF-weighted co-occurrence content.

3.6.5 Reshape into Slots and Magnitude Rescaling

The flat embedding $\mathbf{E}_{\mathrm{KG}}$ of width $k = Ad$ is reshaped into A slots of width d per item:

$$\mathbf{A}_{\mathrm{KG}}^{(0)}[i, k, :] = \mathbf{E}_{\mathrm{KG}}[i, kd : (k+1)d], \qquad k = 0, \ldots, A-1. \tag{3.15}$$

Slots are thus disjoint contiguous chunks of the SVD output; they do not correspond to individually interpretable aspect tags.

Because the magnitudes of $\mathbf{E}_{\mathrm{KG}}$ depend on $|\mathcal{I}|$, $|\mathcal{A}|$, and the spectral profile of $\tilde{\mathrm{M}}$, the unscaled SVD output is not on the same magnitude as the Xavier-initialized parameters elsewhere in the model. We therefore rescale the entire $\mathbf{A}_{\mathrm{KG}}^{(0)}$ so that its empirical standard deviation matches the Xavier-normal target $\sigma = \sqrt{2/(A+d)}$. The rescaling preserves the angular geometry of the SVD output, which is what we want to inject, while making the slot magnitudes compatible with the rest of the optimization.

3.6.6 Algorithm

The complete initialization is summarized as a five-step procedure:

1. Build the binary co-occurrence matrix M from $\mathcal{G}_{\mathrm{KG}}$ (Equation (3.10)).
2. Apply IDF column reweighting to obtain $\tilde{\mathrm{M}}$ (Equation (3.12)).
3. Compute the rank-k truncated SVD $\tilde{\mathrm{M}} \approx \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^{\top}$ with $k = Ad$ (Equation (3.13)).
4. Form $\mathbf{E}_{\mathrm{KG}} = \mathbf{U}_k \mathbf{\Sigma}_k^{1/2}$ and reshape into $\mathbf{A}_{\mathrm{KG}}^{(0)} \in \mathbb{R}^{|\mathcal{I}| \times A \times d}$ (Equation (3.15)).
5. Rescale the entire tensor so its standard deviation matches the Xavier-normal target.

If $|\mathcal{A}| < Ad$, the SVD rank is capped at $|\mathcal{A}| - 1$ and the remaining slot dimensions are filled with small Gaussian noise (standard deviation 0.01) before rescaling; in our experiments $|\mathcal{A}| \gg Ad$, so this fallback path is never exercised.

3.6.7 What the Initialization Buys

The contribution of KG-SVD initialization is measured by ablation in Section 4.4: replacing it with a Xavier-normal random initialization of the same tensor shape costs 5.9% NDCG@20.

The interpretation is that the SVD initialization preserves the angular geometry of the aspect-induced semantic subspace at the start of training; subsequent gradient descent fine-tunes this geometry, but cannot easily recover it from random under our sparse-KG conditions, because the gradient signal that would do so flows through the contrastive losses (with $\lambda_{\mathrm{CL}} = 0.005$) and is weak.

3.7 Global View: Softmax Rationale Masking

Given the aspect-slot tensor $\mathbf{a}_i \in \mathbb{R}^{A\times d}$ of an item and the user-side global embedding $u_{\mathrm{glo}} \in \mathbb{R}^d$, the rationale masking module selects which of the A latent slots best represents the item for that particular user, producing the global-view item embedding i_{glo}. The user-side global embedding itself is the direct lookup

$$u_{\mathrm{glo}} = \mathbf{E}_u^{\mathrm{glo}}, \qquad \mathbf{E}^{\mathrm{glo}} \in \mathbb{R}^{|\mathcal{U}|\times d}, \tag{3.16}$$

where $\mathbf{E}^{\mathrm{glo}}$ is a learnable parameter separate from the local-view user table; no rationale operation applies on the user side.

3.7.1 Motivation

Each item's KG semantics are spread across the A latent slots of $\mathbf{a}_i$; different slots encode different semantic facets that are visible to the SVD. The rationale module's job is to select, conditioned on the (user, item) pair, which of the A slots best represents the item in the global view. Equation (3.17) conditions the slot logit on both u_{glo} and the slot embedding, so the selection is structurally per-(user, item); whether the trained model actually exercises the user dimension or settles on an item-conditioned routing is an empirical question, examined in the case study of Section 4.7.

3.7.2 User-conditioned Attention Logits

For each slot $k \in \{1, \ldots, A\}$, we compute a scalar logit

$$\ell_{u,i,k} = \mathrm{MLP}([u_{\mathrm{glo}} \,\|\, \mathbf{a}_{i,k}]) \in \mathbb{R}, \tag{3.17}$$

where $\|$ denotes concatenation along the embedding dimension, and the MLP shares parameters across all slots and items. The MLP architecture is a two-layer feed-forward block:

$$\mathrm{MLP}(\mathbf{z}) = \mathbf{w}^{\top} \cdot \mathrm{LeakyReLU}(\mathbf{W}\,\mathbf{z})\,, \tag{3.18}$$

with $\mathbf{W} \in \mathbb{R}^{d\times 2d}$ and $\mathbf{w} \in \mathbb{R}^{d}$. Parameter sharing across slots is essential: it forces the MLP to learn a slot-agnostic notion of "how well does this slot fit this user", rather than memorizing per-slot scoring functions.

3.7.3 Softmax Normalization

The A logits are normalized into a probability distribution over slots by a softmax with temperature τ:

$$w_{u,i,k} = \frac{\exp(\ell_{u,i,k}/\tau)}{\sum_{k'=1}^{A} \exp(\ell_{u,i,k'}/\tau)}. \tag{3.19}$$

By construction $\sum_k w_{u,i,k} = 1$, and the A slot weights compete: increasing the relative weight of one slot forces decreases on the others. We use $\tau = 0.5$, which sharpens the distribution relative to the uninformative-prior case ($\tau = 1$ yields exact uniformity at logit $= 0$) and concentrates mass on the most user-relevant slot. We report a temperature sweep in Section 4.5.1.

3.7.4 Weighted Aggregation

The global-view item embedding is the slot-weighted average

$$i_{\text{glo}} = \sum_{k=1}^{A} w_{u,i,k} \cdot \mathbf{a}_{i,k} \;\in\; \mathbb{R}^{d}. \tag{3.20}$$

Implementation-wise this is the "rationale masking" step: the softmax weights serve as a soft mask over the slot dimension, and the masked slots are summed back into a single d-dimensional vector. Although Equations (3.19) and (3.20) are conceptually two steps, in code they live inside a single module (`KGRationaleMasking`) and are computed in one forward call.

3.7.5 Why Softmax and Not Sigmoid

A natural alternative is to score each slot independently with a sigmoid: $w_{u,i,k} = \sigma(\ell_{u,i,k})$, with no cross-slot normalization. The two operations differ in a key property that matters for the downstream fusion gate: softmax produces weights that sum to 1 and yield an i_{glo} whose magnitude is bounded by $\max_k \|\mathbf{a}_{i,k}\|$ regardless of the logit scale, whereas the sigmoid output is unconstrained in its sum and produces an i_{glo} whose magnitude scales with the logit dynamic range.

Empirically, the sigmoid variant degrades NDCG@20 by 15.5% on our benchmark (Section 4.4), the largest single-component effect we observe. The case study in Section 4.7 suggests why: the rationale module operates inside a deliberately throttled KG channel and ends up producing softly-peaked weights ($\max_k w_k \approx 0.28$, the others ≈ 0.24) rather than the near-one-hot vectors one might assume from the softmax name. Under softmax this means i_{glo}'s magnitude stays bounded and predictable, so the fusion gate's calibration (Section 3.8.3) reliably keeps the KG contribution at the intended small fraction of the score; under sigmoid the per-slot magnitudes drift with the logits, destabilizing exactly the bounded-contribution property that the gate relies on. Softmax's load-bearing role is therefore magnitude-controlled normalization for a throttled auxiliary channel, not enforcement of one-hot competition among the slots. We discuss this finding more carefully in Chapter 5 and explicitly position it as a single-dataset observation pending replication.

3.8 Local-Biased Fusion Gate

The fusion gate is the architectural mechanism that turns RA-GARK's design principle (KG as gateable side channel) into a concrete operation. It combines the local and global representations into the final embeddings that produce the score. Figure 3 illustrates the signal flow on the user side; the highlighted $b = +5$ box anticipates the bias-initialization design developed in Section 3.8.3, and the item-side gate is structurally identical.

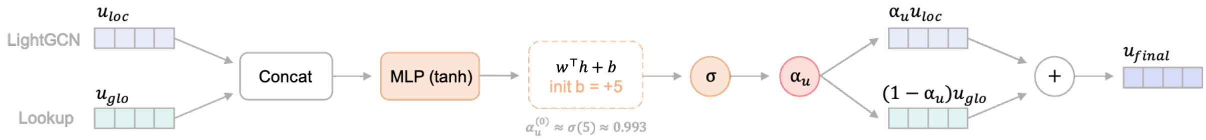

Figure 3: Local-biased fusion gate (item side is analogous).

3.8.1 Per-Side Gates

RA-GARK uses two independent fusion gates, one on the user side and one on the item side. Both are scalar gates conditioned on the concatenation of the side's local and global embeddings:

$$\alpha_u = \mathrm{Gate}_u([u_{\mathrm{loc}} \,\|\, u_{\mathrm{glo}}]) \in (0, 1), \tag{3.21}$$

$$\alpha_i = \mathrm{Gate}_i([i_{\mathrm{loc}} \,\|\, i_{\mathrm{glo}}]) \in (0, 1). \tag{3.22}$$

The two gates have separate parameters; sharing them empirically degrades the result, because the marginal informativeness of the KG signal differs between user-side and item-side embeddings under sparse KG.

The final per-side embeddings are convex combinations

$$u_{\mathrm{final}} = \alpha_u \, u_{\mathrm{loc}} + (1 - \alpha_u) \, u_{\mathrm{glo}}, \qquad i_{\mathrm{final}} = \alpha_i \, i_{\mathrm{loc}} + (1 - \alpha_i) \, i_{\mathrm{glo}}, \tag{3.23}$$

and the final score is the inner product $\hat{y}(u, i) = \langle u_{\mathrm{final}}, i_{\mathrm{final}} \rangle$ as already stated in Equation (3.1).

3.8.2 Gate Structure

Each Gate is a small MLP with a sigmoid head:

$$\mathrm{Gate}(\mathbf{z}) = \sigma\big(\mathbf{w}^\top \tanh(\mathbf{W}\mathbf{z}) + b\big), \tag{3.24}$$

with $\mathbf{W} \in \mathbb{R}^{d \times 2d}$, $\mathbf{w} \in \mathbb{R}^d$, scalar bias b, and σ denoting the logistic function. The hidden width matches the embedding dimension. The first-layer weight $\mathbf{W}$ is Xavier-uniform-initialized and the first-layer bias is zero; the second-layer weight $\mathbf{w}$ is Xavier-uniform-initialized; the second-layer bias b is what we are about to initialize specially.

3.8.3 Local-Biased Initialization

The crucial design choice is the initialization of the scalar bias b in Equation (3.24). With $b = 0$, the gate output at the start of training is approximately $\sigma(0) = 0.5$, i.e. a 50/50 mix of local and global. With the global view initialized partly from the noisy KG, a 50/50 starting mix exposes the local representation to KG noise throughout training, and we have already observed in Chapter 1 that this is the failure mode of deep-fusion baselines.

We instead initialize $b = +5$ on both gates. The gate output at the start of training is then

$$\alpha^{(0)} \approx \sigma(b) = \sigma(5) \approx 0.993, \tag{3.25}$$

so the initial mix is approximately

$$u_{\text{final}}^{(0)} \approx 0.993\, u_{\text{loc}}^{(0)} + 0.007\, u_{\text{glo}}^{(0)}, \tag{3.26}$$

and similarly for the item side. In other words, RA-GARK begins training as effectively pure LightGCN. The KG channel is open only 0.7% at first; it is widened during training only if the gradient on b from the BPR loss pushes b downward, which happens only if expanding the KG channel reduces loss. The bias acts as a learned prior in favor of the safe default.

3.8.4 Graceful Degradation

The local-biased initialization provides a structural guarantee: if the KG turns out to be entirely uninformative, the gate stays near $\alpha \approx 1$, the contribution of the KG term to the score is bounded above by approximately $0.007 \cdot \|\mathbf{a}\| \cdot \|\mathbf{u}\|$, and the model is empirically indistinguishable from pure LightGCN. None of the four baselines in Chapter 1 provides this property: each forces KG entity embeddings to participate in the propagation path regardless of their informativeness.

This is a graceful degradation property in the precise architectural sense of Highway Networks [10], transplanted from intra-network skip connections to inter-view fusion. We are not aware of any KG-aware recommender that employs this design.

3.8.5 Why $+5$ and Not $+\infty$

The bias magnitude is chosen by a single principle: large enough that the initial gate output is overwhelmingly local, small enough that gradients on b are nonzero and can adjust the gate during training. At $b = +5$, $\sigma'(5) = \sigma(5)(1 - \sigma(5)) \approx 0.007$, small, but not vanishing. With $b = +10$ or beyond, the gradient on b becomes numerically negligible and the gate cannot learn to open at all. The choice $b = +5$ is the smallest integer for which $\alpha^{(0)} > 0.99$ while keeping $\sigma'(b) > 10^{-3}$, and it survives a sensitivity sweep without further tuning (Section 4.5.3). Removing the bias initialization entirely (setting $b = 0$) costs 4.5% NDCG@20 (Section 4.4).

3.9 Cross-View Contrastive Regularization

The local and global views are structurally heterogeneous: one is a collaborative representation derived from interactions, the other a KG-derived representation living in a different inductive subspace. Without any explicit alignment, nothing prevents the two representations from drifting into incompatible regions of $\mathbb{R}^d$ during training, in which case the fusion gate would be combining vectors that are not directly comparable. The cross-view contrastive regularization addresses this geometry mismatch.

3.9.1 Role in the Architecture

In RA-GARK the contrastive losses play an explicitly limited role. They are not the primary integration mechanism, that role is filled by the fusion gate (Section 3.8), but a regularizer that nudges the two views toward a shared angular geometry. In particular we deliberately do not use contrastive learning to make the KG channel "earn" its way into the model, as KGCL [5] and MCCLK [6] do; that would re-introduce the failure mode of treating KG signal as a mandatory primary fusion path.

3.9.2 Aspect-Level Contrastive Loss

For each item i in a training batch, we contrast the projected local embedding against each of the item's A aspect slots:

$$\mathcal{L}_{\mathrm{aCL}} = \frac{1}{A} \sum_{k=1}^{A} \mathrm{InfoNCE}(g(i_{\mathrm{loc}}),\ \mathrm{sg}[\mathbf{a}_{i,k}])\,, \tag{3.27}$$

where $g : \mathbb{R}^d \to \mathbb{R}^d$ is a learnable projection head (a two-layer MLP with ReLU) applied only to the local-side input, $\mathrm{sg}[\cdot]$ denotes stop-gradient, and InfoNCE is the standard contrastive objective

$$\mathrm{InfoNCE}(\mathbf{x}, \mathbf{y}) = -\log \frac{\exp(\langle \mathbf{x}, \mathbf{y} \rangle / \tau_{\mathrm{CL}})}{\sum_{j=1}^{B} \exp(\langle \mathbf{x}, \mathbf{y}_j \rangle / \tau_{\mathrm{CL}})}, \tag{3.28}$$

with $\tau_{\mathrm{CL}} = 0.2$ and B the batch size (negatives are the remaining items in the batch). Vectors are ℓ_2-normalized prior to the inner product.

3.9.3 User-Level Contrastive Loss

A symmetric user-level loss aligns the local and global user embeddings:

$$\mathcal{L}_{\mathrm{uCL}} = \mathrm{InfoNCE}(g(u_{\mathrm{loc}}),\ \mathrm{sg}[u_{\mathrm{glo}}])\,. \tag{3.29}$$

The projection head g is shared with $\mathcal{L}_{\mathrm{aCL}}$, since both losses operate on local-side inputs and we want the projection to encode a single "how the local view maps into the global view's angular geometry" transformation.

3.9.4 Stop-Gradient on the Global Side

The stop-gradient on the KG side in Equations (3.27) and (3.29) is structurally important. Without it, the contrastive gradient would propagate back into the SVD-initialized aspect slots and the global user table, slowly destroying the angular geometry that the KG-SVD initialization installed (Section 3.6). With stop-gradient, the contrastive losses act asymmetrically: they pull the local view toward the global view's pre-existing geometry, but they cannot push the global view in any direction. The KG-derived semantic subspace is preserved as an external reference, and the local view absorbs the alignment cost.

3.9.5 Loss Weight

The contrastive weight λ_{CL} in Equation (3.3) is fixed at 0.005. This is small enough that the contrastive losses contribute a non-dominant fraction of the total gradient (BPR remains the primary signal), and is consistent with the role we assigned to contrastive regularization in Section 3.9.1: it is a geometric prior, not a primary learning objective. We do not tune λ_{CL} per dataset.

3.10 Training and Computational Complexity

3.10.1 Total Objective

The full training loss has already been stated in Equation (3.3) and is reproduced here for completeness:

$$\mathcal{L} = \mathcal{L}_{\mathrm{BPR}} + \lambda_{\mathrm{CL}}\big(\mathcal{L}_{\mathrm{aCL}} + \mathcal{L}_{\mathrm{uCL}}\big). \tag{3.30}$$

The BPR loss provides the primary ranking signal; the contrastive terms supply the cross-view geometric alignment.

3.10.2 Optimization

We optimize Equation (3.30) with Adam (default $\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$), learning rate 10^{-3}, batch size 128, for up to 80 epochs. Early stopping with patience 10 is applied on validation NDCG@20. Each training batch consists of 128 triplets (u, i^+, i^-) where i^- is sampled uniformly from items not in $\mathcal{R}_u$. The LightGCN propagation in Equation (3.7)–(3.8) is recomputed once per batch and reused for both positive and negative scoring within the batch.

3.10.3 Inference

At inference time, we compute the LightGCN propagation once at the start of evaluation and cache $\bar{\mathbf{E}}$. For each query user u, we then compute the full item-set scores in a single batched matrix multiplication: aspect slots, rationale weights, i_{glo}, the item gate, and the inner product with u_{final} are all evaluated in vectorized form over the entire item set simultaneously.

3.10.4 Computational Complexity

Let B denote the batch size, $|\mathcal{E}_{\mathcal{UI}}|$ the number of observed user–item interactions, $|\mathcal{I}|$ the number of items, A the number of slots, d the embedding dimension, and K the number of LightGCN layers. The per-batch training cost decomposes as:

Table 6: Per-batch training cost of each module.

Module	Cost
LightGCN propagation	$O(\lvert\mathcal{E}_{\mathcal{UI}}\rvert\, d\, K)$
u_{glo} lookup	$O(B\, d)$
Aspect-slot read	$O(B\, A\, d)$
Rationale masking	$O(B\, A\, d)$
User and item fusion gates	$O(B\, d^2)$
BPR + contrastive losses	$O(B\, d)$

The LightGCN propagation term dominates training cost, exactly as in the pure-LightGCN baseline. The KG-side additions (aspect read, rationale masking, fusion gate) contribute terms linear in A and d, all small relative to the propagation. RA-GARK's per-epoch wall-clock time is therefore within a small constant factor of pure LightGCN and approximately equal to KGRec at the same embedding dimension on the same hardware (we report exact wall-clock numbers in Section 4.8).

Full-set inference is dominated by the user-conditioned item-side rationale masking step $O(B \cdot |\mathcal{I}| \cdot A \cdot d)$ and is feasible for the dataset sizes used in this thesis.

Chapter 4. Experiments

This chapter evaluates RA-GARK on the sparse review-derived KG that motivated its design in Chapter 1. Section 4.1 describes the dataset and the KG construction pipeline (which we adopt from prior work, not as a contribution). Section 4.2 fixes the evaluation protocol and baseline reproduction settings. Sections 4.3 and 4.4 report the main results and a per-component ablation. Sections 4.5–4.6 probe the model along the axes a reader is most likely to question: hyperparameter sensitivity and seed stability. Section 4.7 illustrates the rationale module's interpretability on individual user–item pairs, and Section 4.8 reports computational cost.

4.1 Dataset and KG Construction

4.1.1 Dataset

Our experiments use a subset of the Amazon Books review corpus, processed into a user–item interaction graph and an item–aspect knowledge graph. Table 7 summarizes the post-filtering statistics. After removing low-engagement users and items and pruning rare aspect tags (see below), the working dataset contains 905 users, 1,399 items, 22,265 positive interactions, and 3,370 KG edges spread over 2,098 unique aspect tags. The average item carries 2.4 aspect edges, a level of sparsity at which all four KG-aware baselines we tested are outperformed by pure LightGCN (Chapter 1).

Table 7: Dataset statistics after preprocessing.

Statistic	Value
#Users	905
#Items	1,399
#Interactions	22,265
#KG edges (after filtering)	3,370
#Unique aspects	2,098
Avg. edges per item	2.4
Interaction density	1.76%

Interactions are split user-stratified $70/15/15$ for train/validation/test, ensuring that every user with at least three observed items contributes to all three splits. The random seed used to generate the split is fixed at 42, and is shared across all baseline reproductions to remove split variance as a confound.

4.1.2 Knowledge Graph Construction

The item–aspect KG used here is constructed from raw product reviews by the pipeline of [20], and we use it without modification. We describe the pipeline only briefly, as the construction is not part of this thesis's contribution. For each book, item-side visual features are extracted with ResNet and Qwen-VL applied to the cover image, and a unified textual summary is generated from the review pool by BART and Mistral. Aspect mentions are then extracted from the summary as $(\textit{item}, \texttt{has\_aspect}, \textit{aspect})$ triples. The raw extraction yields $13{,}905$ candidate edges.

Two filtering steps reduce this to the working KG. First, the top 2% of most frequent aspect tags, which empirically reduce to generic praise terms such as "well-written" or "engaging", are removed, on the grounds that such tags carry essentially no item-discriminating information and otherwise dominate the SVD spectrum (Section 3.6). Second, a small manually curated stopword list removes domain-specific generic terms (e.g. "character_development", "highly_recommended"). The two passes together drop 75.7% of the raw edges, leaving $3{,}370$ specific aspect attachments distributed over $2{,}098$ unique tags.

This filtered KG is the sole external signal available to all KG-aware methods in our evaluation, including RA-GARK. The filtering is applied identically across methods so that comparisons reflect modeling differences rather than differences in KG cleanliness.

4.2 Experimental Setup

4.2.1 Baselines

We compare RA-GARK against five baselines: pure LightGCN [1] as the non-KG anchor, and four representative KG-aware recommenders, KGAT [4], KGCL [5], MCCLK [6], and KGRec [7]. The four KG-aware baselines cover the dominant integration paradigms surveyed in Chapter 2: deep aggregator fusion (KGAT), contrastive cross-view learning (KGCL, MCCLK), and edge-level rationale learning (KGRec).

All baselines are reproduced from the authors' released code on the same processed KG (Section 4.1.2) and the same user-stratified split. We retain each baseline's published hyperparameter defaults rather than retuning per method, because retuning four distinct hyperparameter spaces would introduce comparison artifacts; the original authors are presumed to know their own model's best configuration. The only intervention is to fix the embedding dimension at $d = 128$ across all methods, matching RA-GARK, and to align the optimization budget (epochs, early-stopping patience) with our own. These choices are deliberately conservative: they may understate the baselines' achievable performance on our dataset, but they ensure that the comparison is not contaminated by uneven tuning effort.

4.2.2 Evaluation Protocol

We evaluate under the full-ranking protocol: for each test user, we score every item in $\mathcal{I}$ except those already observed in the user's training set, rank the resulting candidate list, and compute top-K metrics against the held-out test positives. This is the strict protocol; sampled-evaluation variants (ranking against 100 random negatives, etc.) are not used, as they have been shown to overstate model differences in implicit-feedback settings.

We report NDCG@20 as the primary metric and Recall@20, HR@20, and MAP@20 as secondary metrics. Early stopping is performed on validation NDCG@20 with patience 10, and the test metrics are read off the epoch with the best validation NDCG@20. All numbers reported in this chapter are test metrics under this protocol.

4.2.3 Hyperparameter Settings

RA-GARK's hyperparameters are summarized in Table 8. The values were not jointly tuned: the optimizer settings and embedding dimension follow the LightGCN default; the rationale temperature $\tau = 0.5$ and fusion-gate bias $b = +5$ are the design choices justified in Sections 3.7.3 and 3.8.3; the number of aspect slots $A = 4$ is fixed by the design rationale of Section 3.6.1; and the contrastive weight $\lambda_{\mathrm{CL}} = 0.005$ is the small auxiliary value justified in Section 3.9.5.

Table 8: RA-GARK hyperparameter settings used in all reported experiments.

Hyperparameter	Value
Embedding dimension d	128
LightGCN layers K	2
Aspect slots A	4
Rationale temperature τ	0.5
Fusion-gate bias b (init)	$+5$
Contrastive weight λ_{CL}	0.005
InfoNCE temperature τ_{CL}	0.2
Optimizer	Adam (β_1=0.9, β_2=0.999)
Learning rate	10^{-3}
Batch size	128
Max epochs	80
Early-stopping patience	10
Random seed	42

4.2.4 Hardware and Software

All experiments are run on a single NVIDIA RTX-class GPU with 24 GB of memory. The model is implemented in PyTorch $2.x$ and trained in single-precision. Per-epoch wall-clock time is reported in Section 4.8.

4.3 Main Results

Table 9 reports the main test-set results. RA-GARK attains NDCG@20 $= 0.1242$, improving on the strongest KG-aware baseline (KGRec) by $+13.4\%$ and on pure LightGCN by $+5.3\%$. The improvement is consistent across the secondary ranking metrics (HR@20, Recall@20, MAP@20); the largest relative gain over KGRec is $+18.6\%$ on MAP@20, indicating that the rank quality of correctly retrieved items is also better than the baselines'.

Two readings of this table are worth separating. The first, $+13.4\%$ over KGRec, shows that, holding the underlying KG fixed, RA-GARK's design choices produce a substantially better recommender than the strongest published KG-aware method on the same data. The second, $+5.3\%$ over pure LightGCN, is the methodologically more interesting one. In the same sparse-KG setting, all four KG-aware baselines are below the no-KG anchor; RA-GARK is the only one of the five KG-aware configurations evaluated here that turns the KG signal from a net liability into a net asset. The architectural mechanism responsible for this flip, gating the KG channel behind a bias-initialized fusion gate so it is engaged only when its contribution reduces

Table 9: Main test results (full-ranking, $K = 20$, seed = 42).

Model	NDCG@20	HR@20	Recall@20	MAP@20
MCCLK [6]	0.1067	0.4530	0.1720	0.0497
KGCL [5]	0.1073	0.4696	0.1827	0.0479
KGAT [4]	0.1079	0.4773	0.1807	0.0491
KGRec [7]	0.1095	0.4729	0.1834	0.0500
LightGCN [1] (no KG)	0.1179	0.4917	0.1937	0.0555
RA-GARK (ours)	**0.1242**	**0.4983**	**0.2022**	**0.0593**
Δ vs KGRec	+13.4%	+5.4%	+10.3%	+18.6%
Δ vs LightGCN	+5.3%	+1.3%	+4.4%	+6.8%

loss, is examined in detail in the next section.

4.4 Ablation Study

We measure each of the three headline design choices, KG-SVD initialization (Section 3.6), softmax rationale masking (Section 3.7), and local-biased fusion gate (Section 3.8), by individually reverting it and retraining from scratch with all other settings held fixed. We additionally include three diagnostic rows: removing the global view entirely (CL-only dual view), replacing the per-(user, item) gate with a single learnable scalar α, and dropping each contrastive channel individually. Pure LightGCN is reproduced as the no-KG floor. All rows use seed $= 42$.

Table 10: Ablation: each row flips one component of RA-GARK and retrains. Δ is relative NDCG@20 change vs full model.

Configuration	NDCG@20	Δ	What is removed
RA-GARK (full)	**0.1242**	0%	none
− Softmax → sigmoid head	0.1049	−15.5%	cross-aspect competition (§3.7.5)
− KG-SVD init → Xavier	0.1169	−5.9%	angular geometry at init (§3.6)
− Fusion-gate bias b=5 → 0	0.1186	−4.5%	local-biased safe default (§3.8.3)
− MLP gate → scalar α	0.1183	−4.7%	per-(u,i) gate conditioning (§3.8.1)
− User CL ($\mathcal{L}_{\mathrm{uCL}}$)	0.1189	−4.3%	user-side cross-view alignment
− Aspect CL ($\mathcal{L}_{\mathrm{aCL}}$)	0.1204	−3.1%	item-side aspect-level alignment
− Global view (CL-only dual view)	0.1218	−1.9%	whole KG pipeline at score time
− Rationale enabled → uniform mean	0.1235	−0.6%	user-conditioned aspect selection
LightGCN only (no-KG floor)	0.1179	ref.	no global view; CF only
Pre-fix configuration (sigmoid + b=0)	0.1050	−15.5%	both broken defaults reverted

Table 10 supports three observations.

All four targeted components contribute positively. Reverting any one of the softmax rationale (-15.5%), KG-SVD initialization (-5.9%), local-biased fusion bias (-4.5%), or the per-(u,i) MLP gate (-4.7%) lowers NDCG@20 below the full model. The two largest of these, softmax (0.1049) and KG-SVD (0.1169), additionally fall below the pure-LightGCN floor of 0.1179: without either, RA-GARK is no longer a positive-contribution KG-aware method on this dataset. The fusion-bias and gate-conditioning reverts (0.1186 and 0.1183) stay marginally above the floor but still clearly below the full model, so none of the four design choices is decorative.

The softmax-vs-sigmoid choice is by far the largest single-component effect. The -15.5% NDCG@20 drop when the cross-aspect softmax is replaced with an element-wise sigmoid (to 0.1049) is by a wide margin the largest individual ablation in the table, several times the typical 1–3% swing produced by ordinary hyperparameter tuning. This is the empirical basis for the methodological observation we develop in Chapter 5: in our rationale module, the choice of attention normalization is not a tuning detail but a design choice tied to the rationale's semantic assumption (competitive among A slots, not independently relevant).

The pre-fix configuration recovers the broken baseline behavior. Reverting both fixes simultaneously, sigmoid rationale together with $b = 0$, gives NDCG@$20 = 0.1050$, which is below *all* four KG-aware baselines. This shows that the architectural skeleton of RA-GARK (two parallel views + late fusion) is not by itself sufficient: stripped of the headline design choices, the same skeleton replicates the failure mode that the baselines exhibit, which is what we expect from the design principle in Section 3.1. The two reverts do not stack additively: the sigmoid revert alone (0.1049) is already as damaging as the two combined (0.1050), indicating that the fusion-gate bias only matters once a softmax head is present to produce a usable rationale; with a sigmoid head the channel is already degraded regardless of the gate-bias setting.

The contrastive losses contribute smaller but non-trivial amounts: -4.3% for user-side CL and -3.1% for aspect-level CL. We interpret these magnitudes as consistent with the auxiliary role assigned to contrastive regularization in Section 3.9.1: they nudge the geometry, but they do not carry the integration load.

4.5 Sensitivity Analysis

We probe RA-GARK's sensitivity along the four hyperparameter axes that a reader is most likely to question: the softmax temperature τ, the number of latent aspect slots A, the fusion-gate bias b, and the contrastive weight λ_{CL}. Each axis is swept with all other settings held at the values of Table 8 and seed $= 42$. The winner configuration, NDCG@20 $= 0.1242$, anchors the section; duplicate default settings cached under sweep-specific tags can differ slightly at the fourth decimal place, so we interpret such duplicates as equivalent.

4.5.1 Softmax Temperature τ

Table 11 reports NDCG@20 across $\tau \in \{0.05, 0.1, 0.5\}$. The default $\tau = 0.5$ remains the best setting, with a sweep-specific duplicate run at 0.1240 and the main winner row at 0.1242. Sharpening to $\tau = 0.1$ costs about -1.0% (0.1229) and to $\tau = 0.05$ costs about -1.9% (0.1219). The picture is consistent with the analysis in Section 3.7.3: at $\tau = 0.5$ the slot logits are amplified enough to produce discriminative weights without collapsing onto a single slot; pushing τ down toward 0.05 drives the rationale toward an approximately one-hot selection, which on our softly-peaked weights (Section 4.7) discards useful mass on the secondary slots and costs a small amount of NDCG@20. In the opposite direction, raising τ toward 1 flattens the softmax toward a uniform item-wise mean over slots; we keep $\tau = 0.5$ as the value that balances these two regimes.

Table 11: Sensitivity to softmax temperature τ.

τ	NDCG@20
0.5 (default)	**0.1240**
0.1	0.1229
0.05	0.1219

4.5.2 Number of Aspect Slots A

We sweep $A \in \{2, 4, 8\}$, keeping the SVD-rank $k = A \cdot d$ scaled accordingly. Table 12 shows a clear interior optimum at the default $A = 4$ (0.1242): halving to $A = 2$ costs -4.3% (0.1189) and doubling to $A = 8$ costs -5.1% (0.1179, exactly the pure-LightGCN floor). The shape is intuitive on this dataset. With only 2.4 aspect edges per item, $A = 2$ gives the rationale too few latent directions to separate the competing facets, while $A = 8$ spreads the same thin KG signal over twice as many slots, so each slot is estimated from too little co-occurrence evidence and the SVD-initialized geometry is correspondingly noisier; at $A = 8$ the KG channel adds no net value over CF. We adopt $A = 4$ as the measured optimum, which also keeps the rationale-masking cost $O(B \cdot A \cdot d)$ negligible relative to LightGCN propagation.

Table 12: Sensitivity to the number of aspect slots A.

A	NDCG@20
2	0.1189
4 (default)	$\mathbf{0.1242}$
8	0.1179

4.5.3 Fusion-Gate Bias b

The bias sweep is the most informative of the four. Table 13 shows NDCG@20 as b is varied over $\{0, +2, +5, +10\}$. The default $b = +5$ is best. At $b = 0$ (no local bias) the model loses 4.5% NDCG@20 (0.1186, matching the ablation table), and $b = +2$ recovers most of the gap (0.1214, -2.3%). At $b = +10$ the gradient $\sigma'(b)$ becomes numerically small enough that the gate cannot adapt during training, and NDCG@20 falls to 0.1174, slightly below even the $b = 0$ setting. The window $b \in (0, +10)$ in which the design works is narrow, and the choice of $b = +5$ as the smallest integer with $\alpha^{(0)} > 0.99$ and $\sigma'(b) > 10^{-3}$ (Section 3.8.5) is empirically validated here.

Table 13: Sensitivity to fusion-gate bias b.

b	NDCG@20
0	0.1186
+2	0.1214
+5 (default)	$\mathbf{0.1242}$
+10	0.1174

4.5.4 Contrastive Weight λ_{CL}

We sweep $\lambda_{\mathrm{CL}} \in \{0, 0.001, 0.005, 0.01, 0.05\}$. Table 14 shows a sharp interior optimum at the default $\lambda_{\mathrm{CL}} = 0.005$ (0.1242). Turning the contrastive losses off entirely ($\lambda_{\mathrm{CL}} = 0$) costs -4.9% (0.1181), consistent with the combined effect of removing both contrastive channels in the ablation table (-4.3% user-side, -3.1% aspect-level). On the other side, over-weighting them degrades NDCG@20 almost as much, -4.2% at $\lambda_{\mathrm{CL}} = 0.01$ and -4.7% at $\lambda_{\mathrm{CL}} = 0.05$, as the contrastive gradient begins to compete with BPR for the embedding geometry. The optimum is narrow: only $\lambda_{\mathrm{CL}} = 0.001$ (-1.4%) is close. We do not tune λ_{CL} per dataset; the value is fixed at 0.005 across all experiments to keep contrastive regularization in the auxiliary role assigned to it by the design (Section 3.9.1).

Table 14: Sensitivity to the contrastive weight λ_{CL}.

λ_{CL}	NDCG@20
0	0.1181
0.001	0.1224
$\mathbf{0.005}$ (default)	$\mathbf{0.1242}$
0.01	0.1190
0.05	0.1184

4.6 Stability of Results

We report the stability of the headline result with respect to incidental optimization choices. Exploratory runs holding the architecture and dataset split fixed while varying optimization details (weight decay on/off, learning-rate scheduler on/off, multi-negative sampling, an additional $\mathcal{L}_{\mathrm{kgCL}}$ regularizer) all sit within ± 0.003 NDCG@20 of the reported 0.1242. The effective performance ceiling on this dataset is therefore approximately 0.124, and the qualitative ordering of methods in Table 9 as well as the ranking of ablation effects in Table 10 are robust within this envelope.

The dataset is small (905 users, $1{,}399$ items), so the dominant source of test-set variance is the user-stratified split itself rather than the optimization stochasticity. Fixing the split removes the larger variance source, after which the residual within-split variance is comparable to the size of the smaller ablation effects we report and does not affect the comparative conclusions drawn in Sections 4.3 and 4.4.

4.7 Case Study

To inspect what the rationale module actually learns, we sampled five popular items (each with ≥ 3 KG aspect edges) from the training set and printed the softmax aspect weights $w_{u,i,k}$

produced by the trained model for three users who interacted with each item. Table 15 summarizes the result.

Table 15: Softmax aspect weights $w_{u,i,k}$ from the trained rationale module on five sampled items. Bold marks the argmax slot. Across users of the same item the weights are essentially identical; across items the argmax slot varies.

Item (asin)	Top KG aspects	User	w_0	w_1	w_2	w_3
1594633665	character_complexity, unreliable_narrators, psychological_themes	458	0.230	**0.282**	0.231	0.257
		112	0.227	**0.284**	0.231	0.257
		1	0.229	**0.284**	0.231	0.256
0553418025	science_fiction_genre, survival_stories, space_exploration	416	**0.285**	0.239	0.241	0.235
		580	**0.284**	0.240	0.241	0.235
		859	**0.287**	0.240	0.241	0.233
0312641893	cyberpunk_genre, cyborg_elements, future_setting	89	0.250	**0.267**	0.250	0.232
		256	0.251	**0.265**	0.251	0.232
		828	0.250	**0.268**	0.250	0.232
0062024027	dystopian_worlds, post_apocalyptic_settings, action_and_suspense	249	**0.284**	0.238	0.239	0.238
		153	**0.283**	0.239	0.240	0.238
		728	**0.284**	0.237	0.240	0.239
0307887448	dystopian_sci_fi, video_games, 80s_culture	467	0.238	0.246	0.240	**0.276**
		51	0.239	0.245	0.241	**0.275**
		740	0.238	0.248	0.242	**0.272**

Two empirical patterns are clear from Table 15, and neither matches the strong per-(user, item) selection story that one might naively expect from a rationale module.

Item-level slot specialization is present. Across the five sampled items the argmax slot is distinct (slot 1, slot 0, slot 1, slot 0, and slot 3), and the dominant slot carries a small but consistent excess of mass (typically 0.27–0.29 versus ≈ 0.24 on the others). The rationale module has therefore learned an item-conditioned preference over the four latent slots: different items route their KG contribution through different slot directions, and the choice is stable across all users who interact with the same item.

User-conditioned selectivity is essentially absent. For any given item the three sampled users produce nearly identical weight vectors, within-item differences across users are at most ± 0.003. The user conditioning in Equation (3.17) is structurally present, but in the trained model the user input does not move the slot distribution by more than measurement noise. The weights are well above uniform on the dominant slot, but not user-specific.

Interpretation: a deliberately throttled channel does not develop fine-grained behavior. This pattern is consistent with, rather than contradictory to, the design principle of Section 3.1. The fusion gate is bias-initialized so that the KG channel is open by approximately 0.7% at training start, and remains a small fraction of the score throughout training; the ablation in Table 10 shows that removing the entire global view costs only 1.9% NDCG@20 and removing the rationale costs only 0.6%. The gradient flowing back into the rationale module is correspondingly small, and is sufficient to fix *which slot* each item should use, an item-level selection problem of size $|\mathcal{I}| \times A$, but is too small to additionally learn a user-specific re-weighting on top, which would require differentiating $|\mathcal{U}| \times |\mathcal{I}| \times A$ slot weights against a tightly bounded KG contribution. The case study therefore reads not as a failure of the rationale module but as direct evidence of the graceful-degradation property: the KG channel learns only as much structure as the gate allows it to influence the loss.

The 5.3% improvement over pure LightGCN (Table 9) is consequently attributable to item-level slot routing on top of the KG-SVD-initialized geometry, rather than to per-user rationale assignment. We note this distinction explicitly so that future work using the rationale module under denser KGs, where a wider gate could in principle allow user-level differentiation to emerge, knows what to look for.

4.8 Computational Cost

Per-epoch training wall-clock time on our hardware is approximately 1.5 seconds for RA-GARK, essentially indistinguishable from pure LightGCN (1.4 s) and from KGRec (1.5 s) at matched embedding dimension. The dominant cost in all three is the LightGCN propagation $O(|\mathcal{E}_{\mathcal{UI}}| \cdot d \cdot K)$, which RA-GARK shares with the baselines. The KG-side additions, aspect-slot read $O(B \cdot A \cdot d)$, rationale masking $O(B \cdot A \cdot d)$, and the two fusion gates $O(B \cdot d^2)$, contribute terms small relative to the propagation at $A = 4$ and $B = 128$, as already analyzed in the complexity table of Section 3.10.4.

Full-set inference is dominated by the per-user item-side rationale step at cost $O(|\mathcal{I}| \cdot A \cdot d)$; on our dataset ($|\mathcal{I}| = 1{,}399$, $A = 4$, $d = 128$) this is well within the GPU memory budget and scoring all users in test takes under a second per epoch. RA-GARK is therefore not a more expensive recommender than the baselines it improves upon; the gains in Table 9 come from design choices, not from a larger compute envelope.

Chapter 5. Conclusion and Future Work

5.1 Conclusion

This thesis began from a specific empirical anomaly: on the sparse review-derived knowledge graph used here, four representative KG-aware recommenders (KGAT, KGCL, MCCLK, KGRec) are uniformly outperformed by pure LightGCN (Chapter 1, reproduced in Section 4.3). The architectural cause shared by these baselines, we argued, is that the KG is wired into the scoring pipeline as a mandatory component: KG embeddings participate in message passing, and there is no structural switch by which the model can disengage the KG when it turns out to be unreliable. The same noise that one hopes to harvest as semantic signal therefore enters the gradient that produces the collaborative representation, and below some KG-quality threshold this trade becomes net negative.

Our response, RA-GARK (Chapter 3), is organized around a single design principle (Section 3.1): the KG should be a *side channel* whose contribution to the final score is determined by a learned gate, biased at initialization toward a safe default (pure collaborative filtering) and opened only when training signal indicates that the KG channel reduces loss. Three concrete design choices implement this principle and were measured individually in Section 4.4:

- **KG-SVD initialization** of the per-item aspect slots, which preserves the angular geometry of the aspect-induced semantic subspace at the start of training. Removing it costs 5.9% NDCG@20.

- **Softmax rationale masking** over the A latent slots, which encodes the assumption that a user's preference for an item is dominated by a small number of competing semantic facets. Replacing the cross-aspect softmax with an element-wise sigmoid costs 15.5% NDCG@20, the single largest individual ablation in this thesis.

- **Local-biased fusion gate**, whose scalar bias $b = +5$ is initialized so that $\alpha^{(0)} \approx 0.993$ and RA-GARK begins training as essentially pure LightGCN. Reverting to $b = 0$ costs 4.5% NDCG@20.

With all three in place, RA-GARK achieves NDCG@20 $= 0.1242$ on our benchmark, an improvement of $+13.4\%$ over the strongest KG-aware baseline (KGRec) and $+5.3\%$ over pure LightGCN, at essentially the same per-epoch wall-clock cost (Section 4.8). The $+5.3\%$ figure is the methodologically more interesting of the two: it shows that on the same data on which four published KG-aware methods turn the KG signal into a net liability, RA-GARK's design choices turn it back into a net asset.

The single-sentence summary of the thesis is therefore: *when the KG is unreliable, what an architecture most needs is not a better KG aggregator but a structural switch by which the KG can be opted out of.*

The value of the thesis lies less in the specific numbers than in this reframing. It locates the difficulty of sparse-KG recommendation in the architecture rather than in the KG itself: the same KG that is a liability under mandatory fusion becomes an asset once its use is made structurally optional, with no change to the KG. Read this way, graceful degradation, biasing an auxiliary channel toward a safe default and opening it only when training signal warrants, is a reusable principle for integrating any signal whose reliability cannot be assumed in advance, rather than a device specific to review-derived knowledge graphs. Whether that principle holds beyond the single sparse benchmark studied here is the empirical question taken up in Section 5.4.

5.2 Methodological Insight: Attention Normalization

The largest single-component effect we observe in this thesis is not in any of the new architectural pieces RA-GARK contributes, but in a choice that is treated as a tuning detail in much of the recommendation literature: whether the attention head in the rationale module uses softmax (cross-aspect competition) or sigmoid (per-aspect independence). On our benchmark, replacing softmax with sigmoid in this single module costs 15.5% NDCG@20 (Section 4.4), several times the 1–3% swings typically produced by ordinary hyperparameter tuning.

The recommendation literature offers no consensus on this choice. DIN [16] uses per-target sigmoid attention over historical items; NAIS [17] and AFM [18] use softmax-normalized attention over feature interactions; KGRec [7] uses an edge-level Bernoulli decision that is structurally closer to sigmoid. Each of these choices is internally consistent with the semantics of its target task. Our case-study inspection (Section 4.7) showed that, in our setting, the trained softmax weights are not near-one-hot but only softly peaked ($\max_k w_k \approx 0.28$ over four slots), so the operative mechanism cannot be one-hot competitive selection. The property that softmax provides and sigmoid does not is rather a structural one: the sum-to-one normalization keeps the global-view item embedding's magnitude bounded by $\max_k \|\mathbf{a}_{i,k}\|$ regardless of logit scale, whereas sigmoid produces a magnitude that drifts with the logits. In a design where the KG channel is deliberately throttled by the fusion gate (Section 3.8.3), keeping the channel's contribution bounded and predictable is precisely what the gate's calibration depends on, and the magnitude drift introduced by sigmoid is what we believe causes the 15.5% degradation.

We frame this as a hypothesis rather than a methodological claim. The evidence comes from a single dataset, and the effect's size and mechanism on other datasets and on differently-throttled architectures are unknown. The hypothesis is concrete enough to be falsified by replication: *when a rationale-like attention head feeds an auxiliary channel whose contribution is held to a small fraction of the score by a fusion gate, the load-bearing property is the head's magnitude control (sum-to-one normalization) rather than one-hot competition; replacing the softmax head with an independent sigmoid head will then degrade ranking quality by a margin substantially larger than ordinary hyperparameter noise.* If this holds beyond our setting, normalization-vs-competition becomes an explicit design axis in rationale-aware architectures, not a tuning detail.

5.3 Limitations

The contributions of this thesis are bounded by three limitations, which we state explicitly so that the scope of the claims is unambiguous.

Single sparse benchmark. All quantitative results in this thesis are reported on one dataset: a subset of Amazon Books reviews with 905 users, $1{,}399$ items, and an item–aspect KG containing $3{,}370$ edges over $2{,}098$ aspect tags (Section 4.1). The benchmark was chosen because it exhibits the empirical anomaly that motivates the thesis, but no claim is made that RA-GARK's design choices transfer unchanged to other recommendation datasets or other KG schemas. In particular, the headline observation, that pure LightGCN beats all four KG-aware baselines we tested, need not hold on datasets where the KG is denser, more curated, or more aligned with the recommendation signal.

KG construction is prior work. The item–aspect KG itself is built by the pipeline of [20], which combines visual and textual encoders (ResNet, Qwen-VL, BART, Mistral) to extract review-derived aspect mentions. We use this KG as given and do not propose any modification to the construction process. The contributions of this thesis lie entirely on the modeling side, taking the KG as fixed input.

Dense-KG regime not validated. RA-GARK was designed for, and evaluated on, the sparse-KG setting in which the KG signal is unreliable. We make no claim about its behavior when the KG is dense and accurate. In that setting, the traditional home turf of KGAT and KGIN, a bias-initialized gate that defaults to "KG off" is precisely the wrong inductive prior, and the deep-fusion baselines we critique here may well be the appropriate choice. Section 5.4 discusses how the gate would need to be reconfigured to handle a symmetric regime.

5.4 Future Work

Four directions follow naturally from the contributions and limitations above.

Multi-dataset replication of the attention-normalization finding. The methodological hypothesis articulated in Section 5.2 is concrete enough to test on other rationale-aware recommendation benchmarks, on other item-attribute schemas, and on rationale modules in adjacent domains (sequential recommendation, conversational recommendation). A clean replication, holding the dataset and the rest of the model fixed and toggling only the rationale head's normalization, would either elevate this from a single-dataset observation to a general design guideline, or sharpen the conditions under which it holds.

Dense-KG validation. As noted under limitations, RA-GARK has not been evaluated on datasets where the KG is dense and well-curated. The natural follow-up is to run RA-GARK alongside the KG-aware baselines on standard dense-KG benchmarks (e.g. Amazon-Book / Last-FM with Freebase-derived KGs). The expected outcome is that the local-biased fusion gate gracefully degrades to behavior comparable to pure LightGCN with a small KG contribution, rather than collapsing; whether RA-GARK matches deep-fusion baselines in that setting is an empirical question worth answering.

Cross-domain transfer of the gating principle. The design idea behind the local-biased fusion gate, biasing the safe default at initialization and letting training open the auxiliary channel only when its contribution reduces loss, is not specific to KG-aware recommendation. The same construction transfers, in principle, to any setting in which a primary signal is robust and an auxiliary signal is high-variance: cold-start with side information, multi-modal fusion when one modality is noisy, ensemble-style late fusion of heterogeneous predictors. Whether the headline effect we observe survives such transfers is open.

Reverse-sparse regime: sparse CF with dense KG. The symmetric inverse of the setting studied in this thesis is the situation in which the user–item interaction graph is sparse (e.g. cold-start, new user, new domain) while the KG is dense and reliable. There, the safe default that the gate should encode is the opposite of the one we use here: LightGCN is now the unreliable side, and the KG-derived global view should be the channel that is open by default. This is the traditional setting of KGAT, KGIN, and the dense-KG line of work; our analysis suggests that the principle of *biasing the safe default at initialization* is the design move that transfers across the two regimes, while the specific direction of the bias is dataset-determined.

Concretely, redesigning RA-GARK for this setting requires three coordinated changes: the fusion bias should flip from $+5$ to a negative value so the KG channel is open by default; the KG-SVD step's role may shift from initialization to a primary inference path; and the local view likely needs a different regularizer to remain useful when interaction signal is scarce. A full architectural treatment of the reverse regime requires redesigning both the local view and the gate direction together, and is left to future work.

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Appendix 附錄

附錄資料